

Winner-Level GEMASTIK Data Mining Playbook

Executive Summary

The strongest public evidence from recent GEMASTIK cycles shows a competition that is **not** decided by leaderboard rank alone. In the official 2023 and 2024 Data Mining rules, the final score was split into **20% preliminary round score, 30% final leaderboard performance, 25% method appropriateness, and 25% presentation quality**, and finalists were given **5 hours** to build a model from a hidden dataset before presenting it onsite. The 2025 official host site and schedule still show the same broad structure—onsite setup, hidden-case solving, and final presentations—while the host statistics page records **258 Data Mining teams** in 2025. For planning purposes, that means a “winner-level” strategy must optimize for **paper quality, method clarity, and presentation discipline** just as much as raw model score. ¹

Recent medal concentration also suggests a real talent-system effect. Public winner announcements show the Data Mining podium repeatedly populated by teams from Universitas Indonesia ², Institut Teknologi Bandung ³, and Telkom University ⁴, while the 2024 national final was hosted at Universitas Negeri Semarang ⁵ and the 2025 edition at Telkom University. This pattern is consistent with repeated institutional training loops, strong mentoring chains, and deliberate preparation for the unusual GEMASTIK format: preliminary research paper first, compressed hidden-case engineering second. ⁶

Across GEMASTIK, KDD Cup, Kaggle, and comparable public challenges, the same methodological theme appears again and again: **winners build pipelines, not single models**. They win by combining careful labeling or weak-label design, leakage-resistant validation, domain-aware feature engineering, hard-example mining, and stagewise ensembling or reranking. The most transferable lesson for GEMASTIK is that “**judgeable robustness**” **beats flashy novelty**: a method that is interpretable, reproducible, and easy to defend under time pressure will often beat a slightly stronger but poorly explained model. ⁷

The highest-upside project space for a student team targeting a win in Indonesia ⁸ is the intersection of **public impact, official/open data availability, and modelability under an 8-week build cycle**. The top six ideas from this report are: wildfire outbreak nowcasting, flood impact nowcasting, dengue early warning, rice-yield anomaly forecasting, earthquake impact prioritization, and food-price shock early warning. All six are feasible with public or quasi-public data, strong baselines, and a paper-first framing that maps well onto GEMASTIK judging. ⁹

GEMASTIK Rules, Finals, and Public Evidence

The official 2023 and 2024 rules are unusually consistent, which is useful because the searchable public evidence for the 2025 division-specific rubric is thinner than for the earlier PDF rulebooks. The 2023 guideline explicitly requires a technical report in the preliminary stage and an onsite final with hidden data, workstation setup, a **5-hour** modeling session, and an in-person presentation. The 2024 guideline preserves the same scoring weights and adds a clearer final deliverables package: model-development documentation, source code, and the team presentation deck. The 2025 host site confirms the competition theme, timetable, number of teams, and the existence of an official handbook download,

but the retrieved indexed snippets do not expose the same division-level rubric detail as the 2023–2024 PDFs. ¹⁰

Cycle	Publicly retrievable evidence	Preliminary phase	Final phase	Scoring evidence	Practical implication
2023	Official guideline PDF; official national winners decree	Technical report on “Data Mining for Advanced and Sustainable Indonesia Development,” including abstract, related work, dataset, method, metrics, experiments, analysis, and conclusion	Hidden dataset; workstation setup on Sep 12; 5-hour model build on Sep 13; PPT documentation and onsite presentation	20/30/25/25 for preliminary score / leaderboard / method / presentation	Plan for a paper-led entry, but rehearse rapid modeling and slide production. ¹¹
2024	Official guideline PDF; official winners decree; university recap from UI	Makalah + similarity test; final upload window also requires final documents	Hidden dataset; workstation setup on Sep 9; 5-hour modeling on Sep 10; presentation Sep 11; ZIP/RAR deliverables with documentation, code, and PPT	20/30/25/25 again	Reproducible code and polished documentation clearly matter more than in many pure leaderboard contests. ¹²
2025	Official host site, competition page, schedule, statistics, ranking page, and UI recap	Official host page lists handbook and general competition materials	Onsite setup on Oct 27 and case solving on Oct 28 per official schedule; final ranking page published	Full division-specific rubric not exposed in retrieved indexed snippets	Use 2023–2024 rubric as the strongest public planning baseline, but assume the same blended emphasis on score + method + presentation. ¹³

Recent winners confirm that the competition is both deep and path-dependent. In 2023, the official decree records **Magnus (ITB)** as gold, **Three Neurons (UI)** as silver, and **Three Wise Monke (ITB)** as bronze. In 2024, the official decree records **Three Outliers (UI)** as gold, **Three Layers (UI)** as silver, and **Barudak Bojongsoang (Telkom University)** as bronze. In 2025, the official host ranking snippet confirms **Three Vectors (UI)** as gold and **Suika (UI)** as silver on the ranking page. ¹⁴

```

xychart-beta
  title "Recorded GEMASTIK Data Mining final leverage points"
  x-axis ["Preliminary score", "Leaderboard", "Method fit", "Presentation"]
  y-axis "Weight (%)" 0 --> 30
  bar [20, 30, 25, 25]

```

The chart above is directly supported for 2023 and 2024. The strategic takeaway is simple: a team that treats GEMASTIK like a Kaggle-only leaderboard sprint is mis-optimizing for roughly **70%** of the score. The real objective is a method that can survive both hidden test data and hostile questioning from judges. ¹⁵

One additional public clue matters. The 2023 ITB gold-medal recap reveals the actual hidden final task that year: finalists received **network log entries** and had to **predict attack type**, with rare classes creating a class-imbalance challenge. That is the clearest public evidence that the final can become a classic “messy hidden tabular/log classification” challenge under severe time pressure. ¹⁶

Winner Case Studies

Public GEMASTIK technical post-mortems are much sparser than the ecosystems around KDD Cup, Kaggle, or DrivenData ¹⁷, so some GEMASTIK rows below explicitly mark missing details rather than guessing. For comparable international contests, the evidence base is much richer: public winning papers, released code, or winner reports exist for many tasks. Where explicit team responsibilities were not published, the “role split” column reflects a **functional inference** from the solution pipeline rather than named responsibilities. ¹⁸

Project	Problem statement	Data and preprocessing	Modeling stack and feature design	Evaluation, compute, role split	Why it won
ILY — GEMASTIK XIV 2021 gold	Detect depressive symptoms from Twitter posts and trigger chatbot outreach for likely cases	Twitter text data; public article indicates significant domain study and consultation beyond pure CS, including outreach to psychology students; exact text cleaning and label protocol not publicly disclosed	Publicly described as AI/NLP-based; exact architecture not disclosed; the clearest published strength is strong NLP understanding and interdisciplinary grounding	Metric, hardware, and ablation not public; likely role split: text/ NLP lead, domain-validation lead, writing/ presentation lead	Won because the problem had clear Indonesian social relevance, strong narrative impact, and domain-grounded NL framing instead of a generic sentiment-analysis paper ¹⁹

Project	Problem statement	Data and preprocessing	Modeling stack and feature design	Evaluation, compute, role split	Why it won
Magnus — GEMASTIK XVI 2023 gold	In the final, predict attack types from hidden network logs; the preliminary topic focused on IoT cybersecurity	Hidden final dataset of network logs; public report explicitly notes rare target classes and the importance of domain knowledge for feature engineering	Exact model not public; published lesson is that feature engineering on logs and handling rare attack classes were decisive	Final operated under a 5-hour build constraint and weighted method/presentation heavily; likely role split: data cleaning + FE, model iteration, slide/documentation	Won because matched a hidden-log tabular task with domain aware feature engineering rather than pure model chasing. ¹⁶
Three Wise Monkey — GEMASTIK XVI 2023 bronze	Pre-GEMASTIK paper on EV impact on air pollution; preliminary GEMASTIK paper on smart energy management using ML for energy-use efficiency; final used hidden onsite data	Public article discloses the research themes but not the final dataset or feature pipeline; notable that the team ramped up within roughly five months of learning ML more seriously	Explicitly described as combining supervised and unsupervised learning for the smart-energy paper; final model details not public	Public evidence shows the same 5-hour final and presentation requirement; likely role split: research framing, modeling, and presentation	Important not because of disclosed architecture, but because shows media winning GEMASTIK work can emerge from strong policy use-case framing plus credible ML toolbox, not only from years of competition experience. ²⁰

Project	Problem statement	Data and preprocessing	Modeling stack and feature design	Evaluation, compute, role split	Why it won
Child Mind Institute Sleep States 1st place	Detect sleep onset and wakeup events from wrist accelerometer time series	Slide transcript reports ~986 MB of CSV/Parquet sensor data with 5-second intervals, core features <code>anglez</code> and <code>enmo</code> , and event annotations per night; CV used 277 series in 5 folds	Five first-level models: CNN+GRU+CNN; CNN+GRU+Transformer+CNN; two LSTM+Unet1d+Unet variants; and 1dCNN+Unet1d+Transformer. A second-level model and custom post-processing converted framewise probabilities into event predictions better matched to AP scoring	Evaluation averaged AP across multiple tolerance windows and across onset/wakeup classes; likely role split: first-level model lead, second-level/meta-model lead, post-processing/validation lead	Won because optimized the actual contest objective at every stage , especially post-processing. This is a direct lesson for GEMASTIK hidden finals where thresholding and formatting can create large score differences. 21
HMS Harmful Brain Activity 1st place — Team Sony	Classify seizures and other harmful EEG patterns	Official challenge provided EEG data and spectrogram representations	Public comparison data identify a huge multi-view ensemble: EEG branch with ConvNeXt Atto ×4 + InceptionNeXt Tiny; spectrogram branch with MaxViT_base ×11 + SwinV2-Tiny ×14	Public comparison table reports a very large ensemble footprint; likely role split: EEG architecture lead, spectrogram lead, ensembling/inference lead	Won through multimodal redundancy of different views of the same physiological signal reduced blind spots. For GEMASTIK, the transferable lesson is not the model size but the value of multi-view representation when data permits it. 22

Project	Problem statement	Data and preprocessing	Modeling stack and feature design	Evaluation, compute, role split	Why it won
Amazon KDD Cup 2024 overall winner	Build an online-shopping assistant for ShopBench: ~20,000 hidden questions over 57 tasks and 5 task types, with only 96 development examples shared	Team built its own training data from multiple public e-commerce datasets and synthetic generation using LLMs; the competition was code-submission-only on Amazon infrastructure	One Qwen2-72B-Instruct model per track; supervised fine-tuning with QLoRA on ~500k examples using 8×A100 80GB; multiple LoRA adapters ensembled; wise-ft for distribution shift; constrained decoding via logits processors; AWQ 4-bit quantization + vLLM for inference	Official paper states phase-2 runtime limits of 20–140 minutes on 4×T4 16GB GPUs; training ~24 hours on 8×A100; likely role split: data synthesis, fine-tuning, inference systems	Won because the team solved the re- hidden-task distribution shift under severe runtime cap . That logic transfers directly to GEMASTIK finals. ²³
Meta KDD Cup 2024 CRAG winner	Build robust RAG systems over web pages and mock knowledge graphs with a 30-second budget per query	Task variants used 5 pages, knowledge-graph APIs, or 50 pages + APIs; HTML was cleaned with BeautifulSoup and chunked with LangChain	Two-path retrieval; retriever/reranker using bge-base-en-v1.5 ; parent-child chunk retrieval with small child chunks and larger parent chunks; tuned LLM; de-hallucination prompts; API regularization for KG retrieval, framed partly like text-to-SQL	Official paper reports three task-winning scores and documents latency-aware chunk choices (e.g., 700/200 baseline); likely role split: retrieval engineer, prompt/KG engineer, evaluation lead	Won because treated latency, hallucination and retrieval granularity first-class modeling problems. In GEMASTIK, the analogue is treating time, memory, and judgeability as part of the model design. ²⁴

Project	Problem statement	Data and preprocessing	Modeling stack and feature design	Evaluation, compute, role split	Why it won
KDD Cup 2024 OAG AQA rank 1	Academic-question-answering track in the OAG challenge	Public GitHub solution used open-source embedding and rerank models plus iterative mining of hard examples; test data from the official challenge portal	Hard-example mining with SFR-Embedding-Mistral , contrastive fine-tuning, top-100 recall for rerank training with SOLAR-10.7B , and a 7-round iterative multipath recall pipeline with rank-average fusion	Single process reached 0.201; 7-round fusion reached 0.223; approximately 40 hours of inference on 8×A100 80GB; full training about 5 days; role split likely divided into retrieval, rerank, and orchestration	Won because aggressively mined difficult negatives and then diversified retrieval rounds before fusing them, very strong template for hidden ranking tasks. ²⁵
KDD Cup 2024 OAG WhoIsWho rank 1	Author name disambiguation in scholarly graphs	Public README describes title-only, author-only, and all-information views of each candidate record	Base LLM was ChatGLM3-6b-32k ; the key idea was to iteratively increase the concentration of positive examples during inference; final merge fused title, author, and full-info variants	Public scores improved from 0.767 and 0.788 single-view runs to 0.8345 final merged score; 8×A100 recommended; likely role split: title-view modeler, author-view modeler, ensemble/merge lead	Won because multiple weak different views of the same entity were more reliable than a single giant model. This is highly transferable structured GEMASTIK entity/linkage tasks. ²⁶

Project	Problem statement	Data and preprocessing	Modeling stack and feature design	Evaluation, compute, role split	Why it won
The BioMassters 1st place	Predict yearly aboveground biomass for forest patches from Sentinel-1 and Sentinel-2 satellite imagery	The challenge used 2,560×2,560 meter forest patches and multi-month satellite data	Public winner summary: combined S1 and S2 into a 15-band 12-month composite , trained a U-Net model, and used test-time augmentation; the wider top-3 ecosystem also featured 3D Swin UNETR and large UNET++ ensembles	Private leaderboard summaries are public, but full compute is not; likely role split: remote-sensing preprocessing, segmentation modeling, inference/TTA	Won because of excellent temporal compositing and geospatial inductive bias. For GEMASTIK it is a model example of converting noisy multi-temporal public data into an analytically stable tensor

27

A useful meta-reading of these cases is that the *winning move* changes by task family, but the *winning discipline* does not. NLP-style tasks reward label and objective design; hidden tabular/log tasks reward fast feature engineering and leakage control; multimodal medical or geospatial tasks reward strong representation choices and aggregation pipelines; retrieval tasks reward hard-negative mining, hierarchy, and latency-aware inference. What stays constant is careful validation and a clear explanation of why the method is appropriate for the data-generating process. 28

Cross-Cutting Patterns and Winner Heuristics

The first cross-cutting pattern is **problem-structure alignment**. Winners rarely begin with “What is the fanciest model I can run?” They start with “What is the exact object the contest rewards?” The Sleep States team optimized event AP with a layered predictor-plus-post-processor pipeline; the Meta KDD winner optimized for 30-second retrieval-grounded answers, not generic LLM eloquence; the OAG AQA winner explicitly targeted the failure regime with hard examples. For GEMASTIK, this means your first design question should be about **what kind of error judges will notice most**: false alarms, missed rare events, unstable explanations, or failure to generalize to hidden slices. 29

The second pattern is **multi-stage modeling**. In winner reports, the final system is usually a stack: retrieval then rerank; coarse model then fine classifier; first-level sequence model then second-level event model; image or signal encoder then tabular/meta learner. Even GEMASTIK public recaps hint at this mindset: what gets remembered is not secret hyperparameters, but domain-informed transformations, feature engineering, or the pairing of supervised and unsupervised analysis. A strong GEMASTIK team should therefore build from a baseline ladder: simple baseline, stronger tabular baseline, task-specific deep model, then optional ensemble or calibration stage. 30

The third pattern is **deliberate treatment of class imbalance, rarity, and distribution shift**. Magnus explicitly faced rare attack classes. Amazon KDD had only 96 visible examples for 57 hidden tasks. OAG AQA improved through hard-example mining. In real Indonesian public-interest problems—floods, outbreaks, price spikes, or landslides—the positive class is also usually rare and the label quality uneven. The practical consequence is that a winner-level project should budget serious time for

thresholding, calibration, hard-negative review, and temporal/region stress testing instead of spending all effort on architecture novelty. ³¹

The fourth pattern is **efficiency as methodology, not implementation afterthought**. Amazon KDD's inference design was constrained by 4×T4 runtime limits. Meta KDD explicitly tuned chunk sizes to fit a 30-second query budget. GEMASTIK finals force a different but equally real constraint: a 5-hour build window and immediate oral defense. A method that is only strong after 40 hours of tuning is strategically inferior to a pipeline that can produce a respectable, reproducible score in 90 minutes and leave enough time for ablation, visualization, and slides. ³²

The fifth pattern is **functional role separation**. Even when public reports do not name responsibilities, the pipelines imply them. Strong teams naturally split into data/feature ownership, model/validation ownership, and documentation/presentation ownership. That division is especially valuable in GEMASTIK because presentation is officially weighted and the final requires not just code but PPT and documentation. ¹⁵

flowchart LR

```
A[Policy or public-service question] --> B[Data audit and legality check]
B --> C[Label design or weak-label strategy]
C --> D[Leakage-safe split and baseline ladder]
D --> E[Task-specific features or representation learning]
E --> F[Advanced model or ensemble]
F --> G[Ablation, calibration, and error analysis]
G --> H[Judge-facing artifacts: report, code, slides, demo]
```

A practical winner heuristic follows directly from the official scoring. Because leaderboard score is only 30% in the documented 2023–2024 rules, the best final-day tactic is usually: **ship one stable baseline early, lock a reproducible validation pipeline, iterate only on a few high-yield ideas, and reserve the last hour for figures and defense**. That method is far more compatible with the competition's blended scoring than a last-minute all-in tuning gamble. ¹⁵

High-Potential Research Ideas

The strongest open-data foundations for GEMASTIK-grade work currently come from BPS ³³, BMKG ³⁴, BNPB ³⁵, NASA ³⁶, and OpenAQ ³⁷, plus official/open geospatial infrastructures such as Copernicus and OpenStreetMap. The table below ranks twelve ideas that fit GEMASTIK's public-impact orientation and are realistic for a student build cycle. The numeric scores are **author estimates** based on impact, data availability, difficulty, compute, and novelty. ³⁸

Idea	Motivation and Indonesia-facing impact	Official/open data sources	Task, baseline, advanced method, evaluation, risks, compute	Scores
Wildfire outbreak nowcasting at village scale	High public value for forestry, health, and regional emergency response	BMKG climate and fire-related climate products; BNPB disaster records; NASA FIRMS active-fire/hotspot feeds; Copernicus Sentinel data; OpenStreetMap exposure layers. 39	Task: rank villages/subdistricts by 1–7 day outbreak risk. Baseline: LightGBM/XGBoost on rainfall deficit, wind, humidity, hotspot history. Advanced: spatiotemporal Transformer or GRU-D + graph exposure model + uncertainty calibration. Eval: PR-AUC, Recall@Top-k, lead-time gain, calibration error. Risks: hotspot labels are noisy; avoid overclaiming causality. Compute: moderate, 1 strong GPU enough.	Impact 5 / Data 4 / Difficulty 4 / Compute 3 / Novelty 4
Flood impact nowcasting with asset-exposure ranking	Directly actionable for local governments, disaster responders, and media rooms	BNPB DIBI and national disaster portal; BMKG weather and forecast products; Copernicus data; OpenStreetMap roads/buildings; Satu Data Indonesia and local open-data catalogs where available. 40	Task: predict district/subdistrict flood event probability and likely impact severity. Baseline: LightGBM on rainfall, antecedent rainfall, river-proxy topography, and exposure counts. Advanced: ConvLSTM or Temporal Fusion Transformer over gridded weather + graph-based exposure propagation. Eval: F1 or PR-AUC for occurrence; MAE/SMAPE for affected-population estimates. Risks: label lag and underreporting. Compute: moderate.	5 / 4 / 4 / 3 / 4

Idea	Motivation and Indonesia-facing impact	Official/open data sources	Task, baseline, advanced method, evaluation, risks, compute	Scores
Rainfall-triggered landslide early warning	Strong relevance in mountainous provinces and transport corridors	BMKG climate and rainfall products; BNPB disaster records; Copernicus terrain/imagery; OpenStreetMap roads and settlements. ⁴¹	Task: predict landslide probability per slope cell or administrative area. Baseline: random forest / LightGBM with rainfall accumulation + terrain proxies. Advanced: geo-GNN or hybrid raster-tabular model combining DEM-derived features and temporal rainfall. Eval: AUROC, PR-AUC, top-k recall on historical events. Risks: sparse labels; potential spatial leakage. Compute: moderate.	5 / 4 / 4 / 3 / 3
District-level dengue early warning	Public-health usefulness is obvious, and BMKG already publishes climate-health related outputs	BMKG climate portal including dengue-related early-warning material; BPS population and regional statistics; Satu Data/local health datasets through data.go.id where district case counts are open. ⁴²	Task: 2–8 week dengue incidence forecast or risk classification. Baseline: SARIMAX + LightGBM with lags and seasonality. Advanced: TFT/DeepAR with rainfall, temperature, population density, sanitation proxies. Eval: MAE/SMAPE for counts; macro-F1 for risk bins; geographic robustness. Risks: health-data completeness, stigma, ecological fallacy. Compute: low-to-moderate.	5 / 3 / 3 / 2 / 4

Idea	Motivation and Indonesia-facing impact	Official/open data sources	Task, baseline, advanced method, evaluation, risks, compute	Scores
Rice-yield anomaly forecasting under climate stress	High relevance to food security and rural policy	BPS agricultural indicators via website/API; BMKG climate and seasonal products; Copernicus Sentinel imagery. ⁴³	Task: predict yield anomaly or risk class by regency/season. Baseline: gradient boosting over lagged climate + historic yields. Advanced: multimodal temporal model combining tabular climate, NDVI-like remote-sensing proxies, and district embeddings. Eval: MAE, R^2 , directional accuracy on anomalies. Risks: alignment of harvest calendars and imagery windows. Compute: moderate.	5 / 4 / 4 / 3 / 4
Food-price shock early warning	Immediate policy relevance for inflation, food affordability, and logistics	BPS statistical products and API; data.go.id and ministry open-data catalogs for food/market series when available. ⁴⁴	Task: anomaly detection or 1–4 week commodity-price forecast. Baseline: Prophet/SARIMAX + LightGBM. Advanced: multivariate transformer with cross-region graph edges and exogenous weather/output signals. Eval: MAPE, extreme-event recall, directional accuracy. Risks: heterogeneous frequencies and regional missingness. Compute: low-to-moderate.	4 / 4 / 3 / 2 / 4

Idea	Motivation and Indonesia-facing impact	Official/open data sources	Task, baseline, advanced method, evaluation, risks, compute	Scores
Urban PM2.5 health-risk forecasting	Strong public communication value, especially in large metros	OpenAQ API and platform; BMKG weather data; optional local open-data inventories via Satu Data. ⁴⁵	Task: nowcast PM2.5 or classify unhealthy episodes. Baseline: LGBM with weather and temporal lags. Advanced: spatiotemporal graph neural network or TFT with sensor-to-sensor transfer learning. Eval: MAE, exceedance-event F1, calibration. Risks: station sparsity, sensor quality heterogeneity. Compute: low.	4 / 3 / 3 / 2 / 3
Rapid earthquake impact prioritization	Very strong response value in the first hour after a quake	BMKG earthquake products; BNPB disaster/exposure geoportal; OpenStreetMap buildings/roads; Copernicus background layers. ⁴⁶	Task: rank affected districts or critical assets immediately after an event. Baseline: rule-based impact index + gradient boosting. Advanced: probabilistic impact model with graph exposure, travel-time degradation, and uncertainty bands. Eval: ranking metrics, MAE on affected population where historical labels exist, scenario backtests. Risks: label scarcity and event rarity. Compute: low-to-moderate.	5 / 4 / 4 / 2 / 4

Idea	Motivation and Indonesia-facing impact	Official/open data sources	Task, baseline, advanced method, evaluation, risks, compute	Scores
Small-boat fishing safety window prediction	Strong maritime livelihood relevance in archipelagic regions	BMKG public weather/geophysics portals; Copernicus Marine services. ⁴⁷	Task: classify safe/unsafe departure windows for coastal zones. Baseline: gradient boosting on wave/wind proxies. Advanced: seq2seq hazard model with geospatial coastal embeddings and uncertainty-aware thresholds. Eval: recall on unsafe windows, false-alarm rate, spatial transfer. Risks: need careful operational definition of “unsafe.” Compute: low.	4 / 3 / 3 / 2 / 4
Health-facility accessibility gap mapping	Strong social-policy framing; excellent for a paper and dashboard combination	BPS regional demographics; Satu Data Indonesia datasets; OpenStreetMap road/facility layers. ⁴⁸	Task: estimate underserved areas and recommend candidate facility placements. Baseline: travel-time analysis + clustering. Advanced: graph optimization or demand-aware facility-location model with learned demand weights. Eval: travel-time reduction, coverage gain, fairness by district type. Risks: not a pure prediction task unless framed clearly. Compute: low.	4 / 4 / 2 / 1 / 3

Idea	Motivation and Indonesia-facing impact	Official/open data sources	Task, baseline, advanced method, evaluation, risks, compute	Scores
Building-resilience and exposure mapping from remote sensing	Strong for disaster resilience, insurance, and planning narratives	Copernicus Data Space; OpenStreetMap; BNPB geospatial layers. ⁴⁹	Task: segmentation/classification of built structures and exposure classes. Baseline: U-Net or Mask R-CNN on small curated areas. Advanced: multimodal remote-sensing encoder + weak labels from OSM polygons + active learning. Eval: IoU/F1, exposure ranking quality, cross-city robustness. Risks: label noise from volunteered map data. Compute: moderate-to-high.	4 / 4 / 4 / 4 / 5
Urban heat exposure ranking for schools and workplaces	Excellent public-health and labor-productivity story	Copernicus land/atmosphere data; BMKG climate products; BPS demographics; OpenStreetMap land use and building layers. ⁵⁰	Task: estimate area-level heat risk and vulnerable-facility exposure. Baseline: gradient boosting on land-cover, density, and climate variables. Advanced: multimodal geospatial model combining remote-sensed heat proxies with facility and demographic embeddings. Eval: rank correlation with observed heat episodes where available, hotspot detection performance. Risks: careful handling of proxy labels. Compute: moderate.	4 / 4 / 3 / 3 / 4

The table favors ideas that can tell a *policy-relevant story* with data sources that judges can trust. That matters because GEMASTIK's preliminary stage is still a technical report competition, not just a benchmark race. A project with moderate novelty but excellent official-source grounding often has higher overall win probability than a more exotic idea with weak data provenance. ¹⁵

Prioritized Shortlist and Roadmaps

The top six ideas below are the ones I would recommend first if the goal is specifically a **winner-level** GEMASTIK submission rather than a broad exploratory portfolio. They balance public value, data realism, technical depth, and defenseability under the official scoring mix. The composite scores are author judgments derived from the evidence above. ⁵¹

Rank	Idea	Upside	Main risk	Feasibility in 8 weeks	Best team profile
Top pick	Wildfire outbreak nowcasting	Strong Indonesia impact, abundant multimodal data, excellent storytelling	Label ambiguity and hotspot false positives	High	tabular/ geospatial + remote-sensing mix
Top pick	Flood impact nowcasting	Very judge-friendly and operationally intuitive	Disaster labels can be delayed and incomplete	High	tabular/ time-series + geospatial
Top pick	Dengue early warning	Public-health impact is compelling and paper-friendly	Case data availability varies by district	Medium-high	time series + careful ethics framing
Strong contender	Rice-yield anomaly forecasting	Food-security narrative and scalable multimodal setting	Crop calendars and label alignment	Medium-high	tabular + remote sensing
Strong contender	Earthquake impact prioritization	Excellent emergency-response value and clear deliverable	Rare-event evaluation and label scarcity	Medium	geospatial + ranking
Strong contender	Food-price shock early warning	Easier compute, easier ablations, strong economic relevance	Data harmonization across sources	High	tabular/ time series

gantt

title Generic eight-week sprint for a winner-level GEMASTIK project

dateFormat YYYY-MM-DD

axisFormat Week %W

section Framing

Problem definition and data audit :a1, 2026-05-05, 7d

section Data

Cleaning, labeling, and split design :a2, after a1, 14d

section Modeling	
Baselines and validation	:a3, after a2, 14d
Advanced methods and ablation	:a4, after a3, 14d
section Packaging	
Error analysis and reproducibility	:a5, after a4, 7d
Report, slides, and rehearsal	:a6, after a5, 7d

Idea	Weeks one and two	Weeks three and four	Weeks five and six	Week seven	Week eight
Wildfire	Lock target geography; ingest BMKG, FIRMS, BNPB; define event labels and exposure grid	Build leakage-safe temporal split; LightGBM baseline; hotspot feature bank	Add spatiotemporal model and uncertainty calibration; do province-wise stress tests	Ablation on hotspot lag windows, climate features, and exposure layers	Final paper figures, demo map, slide rehearsal, “why not false alarms?” defense
Flood	Define impact target: event/ no event vs severity; harmonize BMKG + BNPB + OSM	Strong tabular baseline with rainfall accumulations, drainage/ topography proxies	Add spatial sequence model or graph exposure layer; conduct hidden-slice analyses	Build error atlas by region and event scale	Final package with ranked intervention map and operational case study
Dengue	Secure district case series; align BMKG variables and population denominators	Seasonal/time-lag baseline; judge leakage via forward-only validation	Add TFT/ DeepAR or boosted ensemble; calibrate outbreak thresholds	Ethics section, robustness by province, missing-data sensitivity	Final package emphasizing early-warning lead time and public-health caveats
Rice yield	Choose crop, season, and spatial unit; ingest BPS + BMKG + Sentinel proxies	Build historical yield + seasonal baseline; produce first anomaly map	Add multimodal temporal model; evaluate transfer across provinces	Conduct ablations on imagery windows and climate-only vs multimodal gains	Final package linking predictions to food-security implications

Idea	Weeks one and two	Weeks three and four	Weeks five and six	Week seven	Week eight
Earthquake impact	Define rapid-impact target and historical event set; ingest BMKG + exposure layers	Rule-based and gradient-boosted baseline; construct scenario backtests	Add graph/travel disruption component and uncertainty estimates	Stress-test generalization across event sizes and depths	Final package with “first hour triage” dashboard and judge-friendly scenarios
Food price	Select commodities and markets; harmonize frequencies and locations	Time-series baseline and anomaly detector	Add cross-region graph or transformer model; assess shock recall	Interpretability and ablation by exogenous signal group	Final package centered on inflation/affordability early-warning use cases

If I were coaching a team from zero, I would put the first serious effort into **wildfire**, **flood**, or **food price** because those three offer the best combination of official/open data availability, fast baseline iteration, and strong visual storytelling. If the team already has time-series depth or public-health domain access, **dengue** becomes highly competitive. If the team already knows remote sensing, **rice yield** becomes especially attractive because it can produce a very strong preliminary technical paper and still be engineered into a defensible final-stage narrative. ⁵²

Submission Package and Presentation Strategy

The official GEMASTIK documentation makes the deliverable structure unusually clear. The preliminary stage requires a true technical report with problem framing, related work, datasets, methods, evaluation metrics, experiments, analysis, and conclusion. The 2024 final explicitly requires a package containing model-development documentation, source code, and presentation files, while the 2023 final explicitly required PPT documentation and onsite presentation after the 5-hour build. So the safest winner-level strategy is to design everything from day one as a **submission package**, not as a notebook collection. ¹⁵

Deliverable	Official minimum implied by rules	Winner-level version
Technical report	A complete research-style paper with dataset, methods, experiments, analysis, and conclusion	A paper with a strong Indonesia policy motivation, clear novelty claim, one serious baseline ladder, one ablation section, one explicit limitations/ethics section, and polished figures that can be reused in slides
Experiment evidence	Not stated as a separate artifact, but required to support the paper and final defense	A structured experiment matrix: baseline, stronger baseline, advanced model, ablations on features/splits/losses, plus error slices by region/time/class

Deliverable	Official minimum implied by rules	Winner-level version
Code	Final-source-code deliverable explicitly required in 2024	Reproducible repository with <code>README</code> , <code>environment.yml</code> or <code>requirements.txt</code> , config files, fixed seeds, clear train/infer scripts, and cached outputs for figures
Documentation	Final model-development documentation explicitly required in 2024	Dataset card, pipeline diagram, feature dictionary, training log summary, and a “what to run first” quickstart for judges
Presentation deck	Explicitly required	A defense deck that mirrors scoring: problem value, data credibility, method appropriateness, score, ablations, failure cases, and deployment pathway
Final defense	Onsite presentation	A crisp, role-based defense: one member handles problem/data, one handles model/validation, one handles ablation/limitations and live Q&A

A report outline that maps cleanly onto both the official rules and winner expectations is: **Abstract; Problem and Indonesian relevance; Related work; Data sources and governance; Problem formulation; Baselines; Advanced methods; Experimental setup; Results; Ablations; Error analysis; Explainability and limitations; Ethical/legal considerations; Reproducibility appendix.** The crucial feature is not the headings themselves, but the progression: judges should feel that each design choice followed from the data and the public problem, not from a generic template. ¹⁵

For experiments, the non-negotiables are a **baseline ladder** and a **validation story**. At minimum, run one simple statistical or tree baseline, one strong tabular/sequence baseline, and your advanced method. Then ablate on the elements most likely to matter: temporal split choice, calibration, feature groups, region-specific models versus pooled models, class weighting, thresholding, and ensemble inclusion. The fastest way to lose credibility in a judging room is to present a single strong number without demonstrating why it is believable. That lesson is visible across the public winner literature: Amazon KDD, Meta KDD, OAG, and Sleep States all expose a ladder of improvements rather than a mysterious final artifact. ⁵³

Presentation-wise, GEMASTIK rewards teams that can explain **why the method fits the problem**. The official weighting means a good slide deck should not just show model architecture. It should show, in order, the policy problem, the trustworthiness of the data, the reason the baseline is insufficient, the reason the advanced method is justified, where the model still fails, and what a responsible operational use would look like. In practice, the best 8–10 slides are usually: problem, impact, dataset, split design, baseline table, main model, ablation table, error analysis, deployment/demo mockup, and conclusion. ¹⁵

The final tactical recommendation is to train for the **onsite hidden-case mode**, not just for the preliminary research-paper mode. Run at least two internal simulations where the team has 5 hours to receive a new dataset, clean it, establish a baseline, improve it, and produce a defensible slide deck. That is the closest public-process analogue to what the official 2023 and 2024 rules describe, and it is the most reliable way to convert a good research idea into a medal-capable final performance. ¹⁵

- 1 2 3 10 11 15 32 36 51 <https://pusatprestasinasional.kemendikdasmen.go.id/uploads/lampiran/Pedoman%20GEMASTIK%202023%20-%20revised%2020230509%20%281%29.pdf>
<https://pusatprestasinasional.kemendikdasmen.go.id/uploads/lampiran/Pedoman%20GEMASTIK%202023%20-%20revised%2020230509%20%281%29.pdf>
- 4 49 50 <https://dataspace.copernicus.eu/>
<https://dataspace.copernicus.eu/>
- 5 7 16 30 31 37 <https://itb.ac.id/berita/tiga-mahasiswa-teknik-informatika-itb-raih-juara-1-gemastik-2023-divisi-data-mining/60055>
<https://itb.ac.id/berita/tiga-mahasiswa-teknik-informatika-itb-raih-juara-1-gemastik-2023-divisi-data-mining/60055>
- 6 14 <https://pusatprestasinasional.kemdikbud.go.id/uploads/lampiran/Surat%20Keputusan%20Gemastik%20Nasional.pdf>
<https://pusatprestasinasional.kemdikbud.go.id/uploads/lampiran/Surat%20Keputusan%20Gemastik%20Nasional.pdf>
- 8 39 41 42 52 <https://iklim.bmkg.go.id/id/>
<https://iklim.bmkg.go.id/id/>
- 9 <https://data.go.id/dataset>
<https://data.go.id/dataset>
- 12 <https://pusatprestasinasional.kemendikdasmen.go.id/uploads/lampiran/Pedoman%20GEMASTIK%202024%20-%202003-01.pdf>
<https://pusatprestasinasional.kemendikdasmen.go.id/uploads/lampiran/Pedoman%20GEMASTIK%202024%20-%202003-01.pdf>
- 13 17 34 <https://gemastik18.telkomuniversity.ac.id/dashboard/landing/statistik>
<https://gemastik18.telkomuniversity.ac.id/dashboard/landing/statistik>
- 18 <https://buletingemastik.id/index.php/bg/about>
<https://buletingemastik.id/index.php/bg/about>
- 19 28 <https://itb.ac.id/berita/deteksi-depresi-dari-cuitan-twitter-ily-juara-pertama-gemastik-xiv-penambangan-data/58239>
<https://itb.ac.id/berita/deteksi-depresi-dari-cuitan-twitter-ily-juara-pertama-gemastik-xiv-penambangan-data/58239>
- 20 <https://itb.ac.id/berita/tiga-mahasiswa-teknik-informatika-itb-juara-3-lomba-penambangan-data-gemastik-2023/59942>
<https://itb.ac.id/berita/tiga-mahasiswa-teknik-informatika-itb-juara-3-lomba-penambangan-data-gemastik-2023/59942>
- 21 29 <https://speakerdeck.com/unonao/shui-mian-konpe-1st-place-solution>
<https://speakerdeck.com/unonao/shui-mian-konpe-1st-place-solution>
- 22 <https://www.kaggle.com/competitions/hms-harmful-brain-activity-classification>
<https://www.kaggle.com/competitions/hms-harmful-brain-activity-classification>
- 23 53 <https://ar5iv.org/pdf/2408.04658>
<https://ar5iv.org/pdf/2408.04658>
- 24 <https://openreview.net/pdf?id=oWNPeoP1uC>
<https://openreview.net/pdf?id=oWNPeoP1uC>
- 25 35 <https://github.com/BlackPearl-Lab/KddCup-2024-OAG-Challenge-1st-Solutions/tree/main/AQA>
<https://github.com/BlackPearl-Lab/KddCup-2024-OAG-Challenge-1st-Solutions/tree/main/AQA>
- 26 <https://github.com/BlackPearl-Lab/KddCup-2024-OAG-Challenge-1st-Solutions/tree/main/WhoisWho>
<https://github.com/BlackPearl-Lab/KddCup-2024-OAG-Challenge-1st-Solutions/tree/main/WhoisWho>
- 27 <https://github.com/drivendataorg/the-biomasters>
<https://github.com/drivendataorg/the-biomasters>

33 40 <https://data.bn Timer>

<https://data.bn Timer>

38 <https://data.go.id/>

<https://data.go.id/>

43 44 <https://webapi.bps.go.id/developer>

<https://webapi.bps.go.id/developer>

45 <https://openaq.org/>

<https://openaq.org/>

46 47 <https://www.bmkg.go.id/>

<https://www.bmkg.go.id/>

48 <https://www.bps.go.id/en>

<https://www.bps.go.id/en>